

Packaging Line Camera Quality Specification

Example specification for automated visual inspection and human review routing

Document	QSP-CV-001	Revision	1.0
Owner	Quality Engineering	Effective	13 September 2026
Applies to	Restricted packaging-line camera	Status	Example / not production

1. Purpose and scope

This specification defines camera-visible acceptance rules for a restricted packaging-line inspection frame. The automated system may propose a disposition, but a quality engineer remains responsible for approving extracted rules and resolving review cases.

2. Object and count requirements

2.1. Personnel exclusion

Any person observed inside the restricted camera frame is a critical defect and the inspection result must be REJECT.

2.2. Foreign animal exclusion

Any dog, cat, or bird visible in the restricted camera frame is a major foreign-object defect and the inspection result must be REJECT.

2.3. Maximum restricted-object count

No more than two restricted objects (person, dog, cat, bird, car, bus, or truck) may be present in one image. Three or more restricted objects require REJECT.

2.4. Vehicle escalation

Any car, bus, or truck detection requires HUMAN REVIEW before disposition.

Important: rule extraction creates a draft only. No generated rule may become active until a quality engineer reviews its source evidence and explicitly approves it.

Confidence, measurement, and escalation

Requirements that exceed detector capability must fail safely to human review

3. Confidence handling

3.1. Unconfirmed findings

Unconfirmed detector findings from 30 percent up to but below 65 percent confidence require HUMAN REVIEW. Findings below 30 percent may be ignored as insufficient evidence.

4. Requirements outside the current detector contract

4.1. Scratch measurement

A surface scratch longer than 2 mm requires REJECT.

4.2. Missing component

A missing soldered component requires REJECT.

Requirements 4.1 and 4.2 are mandatory quality requirements, but this example does not define camera calibration, millimetre conversion, a scratch detector, a reference assembly, or component-presence logic. The inspection system must route these requirements to HUMAN REVIEW until the required sensing and detector capabilities are validated.

5. Rule approval and decision authority

5.1. Approval gate

LLM-extracted rules are candidates only and must not be activated automatically.

5.2. Decision authority

The deterministic inspection engine applies approved rules to detector findings. The LLM must not make the final PASS, FAIL, or HUMAN REVIEW decision for an inspected image.

Approval record

Quality engineer	_____
Approval date	_____
Approved policy version	_____